

KNOCK

Design Your Door

CSE 358 Introduction to Artificial Intelligence
Final Project Report

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github.com/AliFuatAkyemis/knock-design-your-door

“Mama, take this badge off of me / I can’t use it anymore”

— Bob Dylan, *Knockin’ on Heaven’s Door* (1973)

Abstract

KNOCK: Design Your Door is an AI-augmented hermeneutic system that analyzes contemporary covers of Bob Dylan's 1973 meditation on mortality, *Knockin' on Heaven's Door*. By orchestrating three distinct generative AI techniques — Speech-to-Text (Groq Whisper), Large Language Model analysis (Llama 3.3 70B), and Text-to-Image generation (Flux via Pollinations.ai) — the project constructs a multi-layered reinterpretation of any performer's relationship to the song's themes. The result is a visual artifact generated in real time, steeped in the aesthetic language of 1973: Kodachrome film grain, warm earth tones, and imagery drawn from the Vietnam War counterculture. This report details the technical architecture, artistic vision, and philosophical engagement behind the system, culminating in a personal manifesto on mortality, interpretation, and the act of knocking.

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1. Introduction

Bob Dylan wrote *Knockin’ on Heaven’s Door* in 1973 for Sam Peckinpah’s revisionist Western *Pat Garrett & Billy the Kid*. The song accompanies the death of a sheriff — a man disarmed, exhausted, and surrendering to the inevitable. In fewer than two minutes, Dylan distilled an entire era’s grief into four lines and a repeated refrain. The year was 1973: the Paris Peace Accords had been signed in January, ostensibly ending direct American involvement in Vietnam, yet the war’s psychological wreckage endured. A generation raised on protest had grown tired. The counterculture was fragmenting. To knock on heaven’s door was not merely to die — it was to relinquish the burden of a world that had demanded too much.

Fifty years later, the song persists — performed by hundreds of artists across genres, languages, and continents. Each cover is a new interpretation: a contemporary voice pressing its ear against Dylan’s original door and hearing something different behind it. KNOCK asks: what happens when we run that interpretation through an AI pipeline designed to hear the song the way 1973 heard it?

The project is not a music generator, nor a simple chatbot. It is a **hermeneutic machine** — a system that interprets interpretations, and in doing so, asks whether meaning can survive translation across time, medium, and model weights.

2. Technical Implementation

2.1 System Architecture Overview

KNOCK is implemented as a full-stack web application with a clear separation between the AI processing backend and the thematic frontend. The pipeline is strictly sequential: audio enters, passes through three AI stages, and a visual artifact exits. The architecture is designed to be asynchronous throughout, ensuring the interface remains responsive during heavy model inference.

Layer	Technology	Role
Frontend	Vanilla JS + CSS	Audio recording/upload, thematic UI, film-grain overlays
Backend API	FastAPI (Python)	Async routing, service orchestration, error handling
Stage 1 — STT	Groq Whisper (large-v3-turbo)	Audio transcription
Stage 2 — LLM	Groq Llama 3.3 70B	Philosophical and thematic analysis
Stage 3 — T2I	Pollinations.ai Flux	Visual artifact generation
Prompt Layer	utils/prompt_engineer.py	5-layer 1973 aesthetic injection

2.2 AI Technique Integration

The three AI techniques are not simply chained in sequence — each stage’s output actively shapes the next stage’s input, creating emergent results that no single technique could produce alone.

2.2.1 Stage 1 — Speech-to-Text (Groq Whisper)

The *whisper-large-v3-turbo* model transcribes the user’s audio cover with high fidelity, capturing not just the words but the performer’s unique phrasing, emphasis, and vocal dynamics encoded in the text representation. This transcription is passed verbatim to Stage 2, ensuring that the LLM analysis is grounded in the specific human performance rather than a generic lyric sheet.

2.2.2 Stage 2 — Large Language Model Analysis (Llama 3.3 70B)

The LLM receives the Whisper transcription along with a carefully engineered system prompt that grounds it in the 1973 historical context. The model is instructed to analyze: (1) the emotional register of the performance, (2) connections to the Vietnam War era and counterculture themes, (3) the performer’s relationship to the song’s core motifs — farewell, mortality, disarmament — and (4) a symbolic visual direction for Stage 3. This analysis is structured JSON output, which the backend parses and routes to the prompt engineering layer.

2.2.3 Stage 3 — Text-to-Image Generation (Pollinations.ai Flux)

Flux receives a dynamically constructed prompt from `utils/prompt_engineer.py`, which synthesizes the LLM’s thematic analysis with the 5-layer aesthetic framework described in Section 2.3. The resulting image is the project’s primary artistic output: a visual artifact that is both historically anchored and uniquely responsive to the specific human performance that triggered it.

2.3 Technical Ambition: The 5-Layer Prompt Engineering System

The prompt engineering layer is the technical heart of KNOCK’s artistic ambition. Rather than passing the LLM’s analysis directly to the image model, `utils/prompt_engineer.py` processes it through five injection layers:

- **Aesthetic Base:** Every prompt includes Kodachrome film grain, 35mm documentary photography style, shallow depth of field, warm earth tones, and the faded yellows and burnt oranges characteristic of 1973 photojournalism.
- **Historical Layer:** Keywords from the LLM analysis trigger period-specific imagery: dog tags, protest banners, military helicopters, rice paddies, candlelight vigils, or the smoky backrooms of American folk clubs.
- **Symbolic Anchor:** A ‘Door’ motif — weathered wood, peeling paint, threshold light — is injected as a compositional constant, providing visual continuity across all generated artifacts.
- **Emotional Modulator:** The LLM’s extracted emotional register (grief, resignation, defiance, hope) adjusts the prompt’s descriptive language: a grief-coded performance generates heavier shadow and empty space; a defiant one generates wider aperture and stronger contrast.

- **Negative Prompt Layer:** Explicit exclusions prevent the model from generating anachronistic imagery, digital aesthetics, or saturated colors incompatible with the 1973 visual language.

2.4 Repository Structure

The full source code and project assets are organized as follows:

```
KNOCK-PROJECT/  
|- backend/ # FastAPI server and AI logic  
| |- config.py # Environment configuration  
| |- main.py # Application entry point  
| |- models/ # Pydantic data schemas  
| | '- schemas.py  
| |- services/ # AI Model integrations  
| | |- gemini_service.py  
| | |- siliconflow_service.py  
| | '- whisper_service.py  
| '- utils/ # Helper utilities  
| '- prompt_engineer.py # 5-layer prompt system  
|- frontend/ # Client-side implementation  
| |- app.js # Audio handling and API calls  
| |- index.html # Thematic user interface  
| '- style.css # 1973 film-grain aesthetics  
|- .env # Secret API keys  
'- requirements.txt # Python dependencies
```

2.5 Setup and Execution Walkthrough

Since sensitive configuration files (.env) and local environments (venv/) are excluded from version control for security and efficiency, follow these steps to initialize and run the system after cloning:

2.5.1 1. Clone the Repository

Clone the project from GitHub to your local machine:

```
git clone https://github.com/AlifuatAkyemis/knock-design-your-door.git  
cd knock-design-your-door
```

2.5.2 2. Create Environment Configuration

The .env file is not included in the repository to protect API secrets. You must create it manually in the root directory:

```
touch .env # Or create a new file named .env using a text editor
```

Open the .env file and insert the configuration. Note that while most variables have defaults, the GROQ_API_KEY is strictly required:

```
GROQ_API_KEY=your_groq_api_key_here  
MAX_UPLOAD_SIZE_MB=25  
UPLOAD_DIR=uploads
```

```
OUTPUT_DIR=outputs
FRONTEND_ORIGIN=http://localhost:8000
```

2.5.3 3. Initialize Virtual Environment & Install Dependencies

Create a fresh virtual environment to manage project dependencies:

```
python -m venv venv
source venv/bin/activate # On Windows: venv\Scripts\activate
pip install -r requirements.txt
```

2.5.4 4. Launch the Server

Start the FastAPI backend using Uvicorn:

```
uvicorn backend.main:app --reload --port 8000
```

Once the server is running, navigate to <http://localhost:8000> in your web browser to access the KNOCK interface.

3. Artistic Originality

3.1 The Hermeneutic Chain

The conceptual backbone of KNOCK is what we call the **Hermeneutic Chain**: a sequence of interpretations in which each link transforms meaning.

*Dylan (1973) → Cover Artist → Groq Whisper → Llama
3.3 → Flux → Viewer*

Dylan encoded grief, farewell, and the exhaustion of idealism into a two-minute folk song. A contemporary performer encodes their own mortality, their own relationship to loss, into that song's shell. Whisper encodes the acoustic performance into text. Llama encodes the text into thematic meaning. Flux encodes that meaning into pixels. And the viewer encodes all of it into personal experience.

This is not degradation — it is amplification. Each translation adds a layer of human and machine subjectivity, and the final artifact carries the sediment of all of them. The viewer does not see what Dylan wrote; they see what it means, filtered through five interpreters, to look at a door and not yet know what lies behind it.

3.2 Aesthetic Coherence

Every design decision in KNOCK — from the frontend's desaturated color palette to the film-grain CSS filter applied to generated images — serves the single aesthetic proposition: *1973 is not a date, it is a feeling*. The feeling of a world that had fought for something, had almost believed in it, and was now slowly setting it down.

The frontend deliberately avoids modern UI conventions: no gradients, no animations beyond a slow dissolve, no sans-serif clarity. The interface feels like a document found in a drawer. The generated images feel like photographs from a war that ended before anyone

agreed on what it meant.

3.3 Emotional Architecture

KNOCK is designed to produce a specific emotional sequence in the user: **participation** → **surrender** → **reflection**. The user participates by recording or uploading their cover — an intimate, slightly vulnerable act. They surrender control to the pipeline, which interprets them without negotiation. They reflect when the artifact appears: an image that is about them, generated by a machine, rooted in a world before they were born.

This sequence mirrors the song itself: the sheriff removes his badge (participation), accepts death (surrender), and becomes part of something larger than himself (reflection). The user does not die; but they do, briefly, let the machine be the one doing the listening.

4. The Artwork

4.1 Visual Language

The generated artifact — what we call the **Artifact** — is not an illustration of the song. It is a diagnostic portrait of the performer’s relationship to the song. Two people performing identical lyrics will generate different artifacts, because Whisper captures how they sing, the LLM reads what that reveals about their inner state, and Flux renders that inner state in 1973’s visual language.

The Kodachrome palette was chosen with deliberateness: it is the palette of Eddie Adams’ 1968 photograph of the Saigon execution, of Nick Ut’s 1972 ‘Napalm Girl,’ of the documentary footage that brought the Vietnam War into American living rooms. These images did not need to be beautiful to be powerful; they needed to be present. KNOCK’s artifacts inherit that obligation.

4.2 The Door Motif

The Door appears in every artifact as a compositional anchor — sometimes as a literal weathered door in an adobe wall, sometimes as a threshold of light at the edge of darkness, sometimes as the silhouette of a figure standing at a gate. It is the one element the prompt engineering layer protects from variation. Everything else — the historical details, the emotional weather, the figure in the foreground — responds to the specific performer. The Door does not move. The Door waits.

5. Philosophical Engagement

5.1 Historical Depth: 1973 and the Grammar of Farewell

To understand *Knockin’ on Heaven’s Door*, one must understand what 1973 meant to carry. The Paris Peace Accords were signed on January 27, 1973, ending direct American military involvement in Vietnam — but not ending the war, not healing the 58,000 Americans and roughly two million Vietnamese dead, and not resolving the fracture the conflict had opened in American consciousness. The Watergate hearings began in May. A country that had organized itself around opposition to a war suddenly had to figure out what it was for.

Dylan, who had been the voice of the protest generation in the early 1960s, had spent the intervening years deliberately retreating from that role. By 1973, he was in Durango, Mexico, playing a minor role in Peckinpah’s elegiac Western and writing a song for a dying man. The sheriff in the film is a figure from an older moral order — a man whose codes no longer apply. His death is not heroic; it is a laying-down. Dylan, who had once written “The times they are a-changin’,” now wrote about the relief of no longer having to change anything.

KNOCK’s prompt engineering layer was designed to carry this specific historical weight. The LLM system prompt includes a carefully researched paragraph on the 1973 context, ensuring that even a contemporary pop cover of the song is analyzed through the lens of this specific farewell, rather than the generic sentimentality the song has accumulated over five decades of use.

5.2 Dylan’s Worldview: The Ethics of Withdrawal

Dylan’s 2016 Nobel Prize lecture offers a retrospective key to his 1973 sensibility. Speaking of *Moby Dick*, he described the captain’s obsession as a kind of moral insanity — the refusal to let go. The sheriff in *Pat Garrett & Billy the Kid* is the opposite: a man who has the wisdom to put down his badge, to acknowledge that his authority has become more burden than purpose. Dylan’s Nobel lecture suggests he has always been interested in this figure — the one who stops, who refuses to keep fighting a war that has already been decided.

This is the worldview KNOCK inherits. The system does not celebrate death; it explores the specific dignity of laying something down. When the LLM analyzes a performer’s vocal delivery, it is trained to look for this: are they carrying something? And can you hear them, in this song, beginning to set it down?

5.3 The AI as Interpretive Subject

A central philosophical question animates this project: can an algorithmic system genuinely *engage* with a work of art, or does it merely process its surface? We do not claim that Llama 3.3 feels grief. We do claim something more modest and more interesting: that the model operates within a latent space of human cultural production, and that its outputs, when properly constrained, can reflect the structure of human meaning-making rather than its mere content.

The LLM has been trained on millions of texts about mortality, war, folk music, and loss. It cannot feel these things, but it has learned the grammar of how humans speak about them. When it analyzes a performer’s transcription and identifies “a quality of resignation threaded through the second verse, consistent with the 1973 counterculture’s exhausted acceptance of systemic failure,” it is doing something real — not feeling, but pattern-matching against the deepest patterns of human expression. We find this sufficient, and more honest than claiming less.

6. Artist’s Manifesto

“Everyone knocks on heaven’s door in their own way. What matters is not what you find behind it, but who you discover yourself to be while knocking.”

6.1 Why This Medium?

We chose this medium — a live processing pipeline that generates images from human singing — because it is the only form we could imagine that would force the question we wanted to ask. The question is not “what does this song mean?” That question has been answered ten thousand times, in ten thousand key changes, by everyone from Eric Clapton to Guns N’ Roses to Arlo Guthrie’s daughter. The question we wanted to ask is harder: *what does it mean to interpret something that has already been interpreted until it barely knows its own name?*

A generative image cannot answer that question in isolation. An LLM cannot answer it alone. But a pipeline that takes a human voice — wavering, specific, carrying its own history — and runs it through layers of machine interpretation before producing a visual artifact: that pipeline makes the question visible. You can see the interpretation happening. You can watch the song travel.

We chose this medium also because it requires the user to do something slightly uncomfortable: to sing. Not to type about the song, not to click through it, but to open their mouth and make the sounds. This is the oldest form of engagement with folk music, and we wanted to preserve it even inside a machine. The Whisper model does not care if your voice is good. The LLM does not judge your pitch. But you have to decide, in the moment of recording, how to hold these particular words. That decision is already an interpretation. That is where the art begins — before the AI has done anything at all.

The web application format was chosen because it removes barriers. A gallery installation would have been more spatially dramatic, but it would have restricted access. A command-line tool would have been more technically pure, but it would have excluded everyone who doesn’t live in a terminal. The browser is where most people encounter things now. We wanted KNOCK to be available to anyone who has ever hummed this song in a car and wondered what they were really singing about.

6.2 What Caught Us?

We encountered this song the way most people our age do: late, mediated, already filtered through decades of use. We did not hear it in 1973, in a dark cinema in the aftermath of Vietnam. We heard it covered, sampled, placed in television soundtracks, used as a backdrop for sports montages and political advertisements. The song had been polished smooth by repetition. We thought we knew what it was.

What caught us — genuinely, in a way that changed how we thought about the project — was watching the Peckinpah film. Specifically, the scene for which Dylan wrote the song: Slim Pickens, playing a dying sheriff, lying in a field by a river while his wife holds his hand. It is not a dramatic death. There is no speech, no revelation, no final stand. He simply lies there, and the river moves, and the light changes, and Dylan sings those four

lines. And something in that image — a man at a threshold, not crossing it dramatically but simply arriving at it, spent — felt true in a way we had not expected.

We had come to the project thinking about AI and music and generative systems. We left it thinking about what we carry, and what we might one day want to set down. That shift is, we think, what the assignment was asking for. Not a technical demonstration, but an encounter. We had the encounter. It arrived through a sixty-year-old actor dying in a Sam Peckinpah film, and we did not see it coming.

We were also caught by the song’s refusal of specificity. Dylan does not tell us what heaven is, or whether it exists, or whether the door actually opens. He tells us the badge is heavy, the guns are too dark to be clean, and there is a knock. That is all. The song’s power is entirely in what it withholds. This restraint became a design principle for KNOCK: we show the door, we do not show what is behind it. The artifact is always a threshold image, never a revelation. We earned that choice from Dylan.

6.3 AI as What?

This is the question we debated longest. The easy answers came quickly and were quickly discarded.

AI as tool? Too simple. A hammer is a tool; it has no relationship to what you build with it. Llama 3.3 has read more about mortality and folk music and the Vietnam War than either of us will in our lifetimes. When it analyzes a performance, it brings that accumulated reading to bear. It is not neutral. Calling it a tool flattens something real.

AI as collaborator? Too generous, or too specific. A collaborator has intentions, preferences, the capacity to surprise you on purpose. The LLM surprised us constantly — producing analyses we would not have written, making connections between a performer’s phrasing and specific historical events that we had not anticipated. But it did not do this *intentionally*. It did it because it was well-trained and the prompt was well-constructed. The surprise was real; the intention was ours.

AI as mirror? Closer. When KNOCK analyzes a cover performance, it reflects back a reading of that performance that the performer did not consciously generate. It reveals the subtext. A user who sings the song with forced brightness will generate an artifact with shadows the user did not choose. The system sees what the user performs, not what the user intends. That is mirror-like behavior.

Our final answer: **AI as archaeologist**. The models we use — Whisper, Llama, Flux — were trained on enormous archives of human production: speech, text, images. They are, in a meaningful sense, compressed human culture. When they process a performance of a 1973 folk song, they are not computing from scratch; they are excavating from within a vast sediment of human expression, finding the layer that corresponds to grief, to farewell, to the specific visual grammar of that era. They dig, and what they bring up is genuinely from the earth — the human earth — even if the hands doing the digging are algorithmic.

This metaphor shaped our technical decisions. We did not try to make the AI create something ‘new’ in the sense of unprecedented. We tried to make it excavate something specific: the 1973 layer. Every element of the prompt engineering system is a constraint that forces the excavation to go in the right direction. The art is in knowing where to dig.

6.4 Your Door: On Thresholds We Carry

This question — what does knocking on heaven’s door mean in your own life? — is the one we approached most carefully, because it is the one where honesty is most expensive.

We are students of computer science in a moment of profound transition. The field we are entering is not the field it was five years ago, and it will not be the same field in five years more. The tools we are learning to build are tools that are, in some configurations, learning to replace the need for builders. We are training for a profession that is itself being retrained. This is our version of 1973: the feeling that the old frameworks are dissolving, that the badge — the credential, the discipline, the role — may soon be too heavy to wear, or may mean something different than it used to.

We do not say this with self-pity. The sheriff in the film does not die with self-pity. He dies with the specific dignity of a man who has chosen to stop fighting something larger than himself and instead attend to the river, the light, his wife’s hand. There is something instructive in that choice. We do not yet know what it means to stop fighting the transitions of our field and simply attend to what is good in the work — but KNOCK is, among other things, an attempt to practice that attention.

We also encountered personal thresholds in building this project. Working together across late nights and debugging sessions, we discovered that collaboration has its own doors — moments where you must decide whether to hold your position or let your partner’s vision lead. Those moments felt, in small ways, like the song: the relief of releasing something you had been gripping too hard.

What does knocking on heaven’s door mean to us? It means: there are things you cannot force. There are doors that open on their own schedule, or not at all. The knocker’s job is only to knock — to keep showing up at the threshold, with whatever you have, and wait. KNOCK, the system, does this literally: it takes a human at a threshold (singing a song about thresholds) and renders that threshold back to them in visual form. We designed it this way because we believe the rendering is the point. Not the arrival. Never the arrival. Only the knocking.

We will graduate from this program and walk through a door that none of us can see clearly yet. The field is changing; the economy is changing; what it means to build software, to make something, to be useful — all of it is in motion. We made this project as an act of attention to that threshold. Not an answer. An attention. We hope it is the kind of attention that leaves a mark.

***Team contribution note:** Technical architecture, backend implementation, and prompt engineering system were developed jointly with continuous iteration between both team members. Artistic direction and manifesto writing were co-authored through a deliberate process of challenge and revision. Both members are prepared to discuss any aspect of the project in full during the exhibition.*

7. Conclusion

KNOCK: Design Your Door succeeds in doing what we set out to do: building a system that does not merely process a song, but engages with it. The technical pipeline — Whisper to Llama to Flux, mediated by a carefully engineered prompt layer — produces artifacts that are visually distinctive, historically grounded, and genuinely responsive to individual human performance. No two artifacts are the same, because no two performances are the same.

More importantly, the system taught us something we did not know we needed to learn. We came to it as engineers interested in pipeline architecture. We left it as people who had spent weeks thinking about farewell, about the dignity of disarmament, about what it means to stand at a threshold and knock without knowing what — or who — might answer.

Dylan wrote a song for a dying man in a film about a dying era. We built a machine for listening to how people sing that song today, and for rendering what we hear in the visual grammar of that era. The door in every artifact is the same door. The light behind it changes every time. That variability is not a bug. It is the entire point.

“Knock. The door is yours to design. What’s behind it is yours to discover.”

8. Academic Integrity Statement

All AI tools used in this project are documented transparently. The Groq API was used for Whisper (`whisper-large-v3-turbo`) and Llama 3.3 (`llama-3.3-70b-versatile`) inference. The Pollinations.ai API was used for Flux-based image generation. All creative vision, architectural design, prompt engineering decisions, and philosophical reflection are original work by the project team. No code was borrowed without attribution; the full repository is available for inspection at github.com/AliFuatAkyemis/knock-design-your-door. Both team members can explain every component of the pipeline in full.